

# Software Requirements Specification (SRS)

AMAHIRWE Project

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ALCHE

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| Name           | Date         | Reason For Changes                                                                                            | Version |
|----------------|--------------|---------------------------------------------------------------------------------------------------------------|---------|
| Lisette Mukiza | 13 July 2026 | Initial draft                                                                                                 | 1.0     |
| Lisette Mukiza | 30 July 2026 | Edited the Appendix B section by adding the 4 UML diagrams: use case, sequence, class, and flowchart diagrams | 2.0     |
|                |              |                                                                                                               |         |

# 1. Introduction

## 1.1 Purpose

This document explains the requirements of Amahirwe software, its version being 1.0. It is going to be a mobile application assisting students in rural public schools in revealing their talents and being put in touch with mentors, teachers, and people providing opportunities, like scholarships and competitions. The SRS covers the whole first launch: student profiles about talents, mentor matching, the teacher's dashboard, and the feed of opportunities. The multi-country expansion includes later releases after testing and modification in Rwandan pilot.

## 1.2 Document Conventions

In this document, standard headings and numbered requirements are employed. There's no unusual use of typography here. Functional requirements are labelled with the FR-x format and put under the feature they relate to, and all non-functional requirements have the NFR-x format. While requirements may not have fixed priority in this version, the level of priority will be determined sprint by sprint based on student and teacher feedback. The agile process

applied here makes the prioritization of the requirements easier during the overall software development process.

## **1.3 Intended Audience and Reading Suggestions**

This SRS was created within a specific context of three main target audiences: instructors of this course, other developers, including myself participating in future sprints and, lastly, potential partners such as ALCHE contacts, TEDxALCHE organisers or school authorities in Rwanda who may be interested in the functioning of the platform. Readers interested in the functioning of the platform should consult Section 2 (Overall Description) and Section 4 (System Features). Readers interested in construction and evaluation of the platform should pay attention to Section 3 (External Interface Requirements) and Section 5 (Nonfunctional Requirements).

## **1.4 Product Scope**

Amahirwe assists students in rural areas of Rwanda discover their abilities and give them the opportunity to use their skills. Amahirwe should not be regarded as a replacement for conventional education or as a regular e-learning platform. Its goal is different from that since it aims at turning the hidden talent into something that can be recognized and transformed into a genuine connection. While the initial version of the project will be implemented in Rwanda, in the Eastern Province, the design of the system allows the expansion to other countries without having to change the architecture. The long-term objective of the ALCHE network is to implement it in different countries across Africa.

# **2. Overall Description**

## **2.1 Product Perspective**

Amahirwe is an entirely new product. It is not a replacement or upgrade to an existing system and to the best of my knowledge, there is no such platform that focuses on

talent-to-opportunity rate specifically targeting rural students in Rwanda. Available solutions are either general e-learning applications or career services working in Kigali only. Therefore, it will have to cooperate with existing tools used by schools or families rather than break away from what they use already. For instance, WhatsApp is commonly used by Rwandan teachers and parents for communication, so Amahirwe will work in cooperation with what they are already using without getting into competition for their attention.

## **2.2 Product Functions**

In general terms, the platform should be able to perform the following functions:

- Create student talent profiles by using a simple evaluation process during which students take an examination offline.
- When Internet connection is established, the profile is automatically uploaded to the server.
- Connect student profiles with mentors and providers of opportunities according to shared interests and talents.
- Give teachers a tool for tracking students' information as well as for advocating appropriate matches for students.
- Provide relevant information on scholarships, competitions, and internships to students and teachers.
- Monitor students' activity and award them with certificates/badges upon achieving certain milestones.
- Inform students' parents in one of the three languages (Kinyarwanda, English, or French) about important changes in students' profiles.

## **2.3 User Classes and Characteristics**

The users in this case are students who are between the ages of 12 and 19 and are learners in rural public schools with a varying degree of technological know-how because the system assumes the use of their platform by the first-time smartphone users with no prior knowledge of using a smartphone. The teachers serve as credible intermediaries necessary for the students to confirm the validity of their profile as students cannot communicate directly with

adults without an educator being a part of this process. The volunteers that can provide mentorship are adults who are vetted and usually former members of ALCHE, TEDxALCHE, and other members of the diaspora community willing to work on a loose mentoring format. Opportunity providers are schools, NGOs, or organizations posting real opportunities. Parents only have access to the summary but not to the full version of the platform.

## **2.4 Operating Environment**

The web application will be a progressive web application and will be fully operable within a regular mobile browser environment, suitable for use on low-cost Android-based devices, which is the type of device most familiar to the users of the system. The backend is using Node.js platform and PostgreSQL database, hosted on a low-cost cloud provider with access from East Africa. The system should be able to operate reasonably well under 2G/3G connectivity and cope even with periods of disconnection from a mobile network.

## **2.5 Design and Implementation Constraints**

- It is necessary for it to function on basic Android devices and also older operating systems as these are still widely used in remote regions of Rwanda.
- It is necessary for it to use the minimum resources in terms of data because price of 1GB of data can take a good part of the monthly expenses of a poor family in the mentioned country.
- It is necessary for this function to obey use of Kinyarwanda in a principal manner, not just as an additional language.
- It is necessary for it to operate with the possibilities of the WhatsApp Platform by means of special notifications and marketing functions.
- It is necessary for it to function on the premise of the restricted budget, as this is a pilot project funded only by internal sources.

## **2.6 User Documentation**

Documentation associated with this release is very minimal: a brief in-app onboarding walkthrough appears the first time a student, teacher, or mentor logs in, supporting all three languages used in this release. It is also accompanied by a printed guideline to be kept by teachers at school, that covers everything that may be needed by students who do not own a phone. The help line is provided through WhatsApp and monitored by the person in charge during the pilot, replacing the help desk for the time being.

## **2.7 Assumptions and Dependencies**

- Assumes that at least one individual from each involved pilot school is either the owner of a smartphone device or has regular access to one.
- Assumes that there is a partnership with an NGO or a telecom provider for the subsidisation of mobile data costs during the pilot stage of the project, as the project has not determined methods through which to solve this issue itself.
- Assumes that the WhatsApp Business API will remain available and that a basic SMS gateway will be implemented in case it is not.
- Assumes a positive attitude from school management and, when needed, the Rwanda Education Board towards the implementation of the pilot project.

# **3. External Interface Requirements**

## **3.1 User Interfaces**

The design of the interface utilizes large, icon-based buttons that also include brief texts. Long texts would make it difficult to use data-dependent applications as first-time users may have difficulty deciphering information. If a user makes a mistake, the application will play a recorded message in Kinyarwanda for clarification. Each screen has a navigation button and a language-switching button (Kinyarwanda, French, and English). The design of the teacher dashboard is simple. It consists of students' names, their current status, and the list of mentees assigned to the teacher.

## **3.2 Hardware Interfaces**

The platform concentrates on Android phones around market average with fully functional cameras, although a student will not be specifically asked for their profile picture. However, they may need to record a short video or voice message that clearly shows their skills and abilities in assessment. No other special equipment is needed.

## **3.3 Software Interfaces**

The system combines the WhatsApp Business API for sending messages to users and notifications on matches, PostgreSQL database for saving profiles and match data, and an SMS gateway that is most likely Africa's Talking for any household that does not have a device compatible with the web application. Node.js framework is used to create the backend of the system that has been designed to use REST API by the web app frontend.

## **3.4 Communications Interfaces**

All the communications between the client and the server take place through HTTPS. Since connectivity is not constant, the application allows local queuing of operations like completed test or updated profile such that when a new connection occurs it transmits everything at once rather than requiring a permanent connection for functioning.

## 4. Requirement Specification

### 4.1 Talent Discovery and Student Profile

This is the entry point for every student. A short, offline-capable assessment (roughly 10 to 15 questions, mixing text, images, and optional voice notes) helps a student surface an area of strength, whether that is visual art, sport, technology, public speaking, or something else, without requiring them to already know what they are good at. A teacher in a rural school should be able to recommend a student they've noticed, even if that student is shy about registering themselves; the profile can be started by a teacher and later confirmed by the student.

| Req ID | Requirement                        | Description                                                                                                                                  |
|--------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1.1 | Complete talent assessment offline | Allow a student to complete the talent-discovery assessment fully offline, with no internet connection required to start or finish it.       |
| FR-1.2 | Create a student profile           | Allow a student or supervising teacher to create a basic student profile including name, age range, school, and preferred language.          |
| FR-1.3 | Queue and sync profile data        | Store completed assessments and profile edits locally on the device and sync them automatically to the server once connectivity is detected. |

### 4.2 Mentor Matching

Once a profile shows a clear talent area, the system should be able to suggest a suitable mentor or opportunity provider from the verified pool, rather than leaving the student to search on their own. Matching in this first release is intentionally simple, based on talent area and language, rather than a complex recommendation model, since the priority is getting a working match in front of real students quickly and improving the matching logic in later sprints based on what actually happens.

| Req ID | Requirement                       | Description                                                                                                  |
|--------|-----------------------------------|--------------------------------------------------------------------------------------------------------------|
| FR-2.1 | Generate mentor match suggestions | Suggest one or more verified mentors to a student based on the student's talent area and preferred language. |

|        |                                          |                                                                                                                                       |
|--------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| FR-2.2 | Require teacher approval before contact  | Require a teacher or administrator to approve a suggested match before any contact details are shared between a student and a mentor. |
| FR-2.3 | Notify both parties of a confirmed match | Send a WhatsApp or SMS notification to the mentor and to the student's teacher once a match is approved.                              |

### 4.3 Teacher Dashboard

Teachers need a way to see, at a glance, which of their students have completed a profile, which have a pending match awaiting approval, and which need a nudge to finish their assessment. This dashboard is the main reason a teacher would keep coming back to the platform, so it needs to answer that question quickly rather than burying it under menus.

| Req ID | Requirement                       | Description                                                                                            |
|--------|-----------------------------------|--------------------------------------------------------------------------------------------------------|
| FR-3.1 | View student profile status       | Allow a teacher to view the profile completion status of every student they supervise on the platform. |
| FR-3.2 | Approve or decline mentor matches | Allow a teacher to approve or decline a suggested mentor match for any of their students.              |

## 4.4 Opportunities Feed

Beyond one-to-one mentoring, students should be able to see scholarships, competitions, and internships relevant to their talent area, posted by verified opportunity providers. This feed is what turns a single mentor connection into a wider set of possibilities, and it also gives opportunity providers a reason to keep using the platform beyond a single match.

| Req ID | Requirement                         | Description                                                                                                                                             |
|--------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-4.1 | Post an opportunity                 | Allow a verified opportunity provider to post a scholarship, competition, or internship with a title, description, deadline, and eligible talent areas. |
| FR-4.2 | Filter opportunities by talent area | Allow a student or teacher to view opportunities filtered by the student's talent area and location.                                                    |

## 4.5 Offline Sync and Low-Connectivity Support

Since connectivity is the central constraint of this whole project, offline behaviour is treated as its own feature rather than an edge case bolted onto the others. Every action a student or teacher takes while offline needs to be saved locally and synced later without data loss or duplication.

| Req ID | Requirement                                | Description                                                                                                                                          |
|--------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-5.1 | Detect connectivity and sync automatically | Detect when a device regains connectivity and automatically sync any locally queued data without requiring the user to trigger it manually.          |
| FR-5.2 | Prevent duplicate or lost records on sync  | Ensure that a sync interruption, such as a connection dropping mid-upload, does not create duplicate profiles or cause loss of already-entered data. |

## 4.6 Multilingual and Accessibility Support

Kinyarwanda is the first language of the great majority of intended users, so it cannot be treated as a translation layered on top of an English-first design; text, voice notes, and notifications all need to work natively in Kinyarwanda, with English and French available for teachers, mentors, or partners who prefer them.

| Req ID | Requirement                                | Description                                                                                                                    |
|--------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| FR-6.1 | Switch interface language                  | Allow any user to switch the interface language between Kinyarwanda, English, and French at any time.                          |
| FR-6.2 | Provide voice-note guidance for assessment | Provide an optional Kinyarwanda voice note explaining each talent-assessment question for students who find reading difficult. |

## 5. Other Nonfunctional Requirements

The table below summarises the non-functional requirements for this release; the subsections that follow explain the reasoning behind the main categories.

| Requirement Type | Req ID | Description                                                                                                                                                           |
|------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance      | NFR-1  | The app must load its core screens within 5 seconds on a typical 3G-equivalent connection, and the assessment must remain fully usable with no connection at all.     |
| Safety           | NFR-2  | No direct messaging channel exists between a mentor and a student without a teacher or administrator in the loop, given that primary users are minors.                |
| Security         | NFR-3  | All student data is encrypted in transit and at rest, and mentors and opportunity providers must be verified by an administrator before they can appear in any match. |
| Usability        | NFR-4  | The interface must be usable by a first-time smartphone user with limited reading confidence, supported by icons and optional voice guidance.                         |

|              |       |                                                                                                                                         |
|--------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Auditability | NFR-5 | The system must log every mentor match decision, including who approved it and when, so the pilot's outcomes can be reviewed afterward. |
| Reliability  | NFR-6 | Locally queued offline data must survive an app restart or device reboot without loss before it has synced.                             |
| Localization | NFR-7 | The system must fully support Kinyarwanda, English, and French across all screens and notifications, not only on marketing pages.       |

## 5.1 Performance Requirements

Because most students will be on 2G or 3G connections at best, and often none at all, performance here is really about graceful degradation rather than raw speed. The assessment and profile screens must work fully offline; only the sync step and the opportunities feed genuinely require a connection, and both should recover cleanly once one becomes available.

## 5.2 Safety Requirements

Since the primary users are minors, safety takes priority over convenience wherever the two are in tension. No mentor gains direct, unsupervised contact with a student; a teacher or administrator approves every match, and communication in the pilot phase is expected to run through a supervised group channel rather than private messaging.

## 5.3 Security Requirements

Student data, including talent assessments and contact details, is treated as sensitive and encrypted both in transit and at rest. Mentors and opportunity providers go through a manual verification step by an administrator before their profiles can be matched to any student, since the trust of teachers and parents in this platform depends entirely on who is allowed onto it.

## 5.4 Software Quality Attributes

Usability and offline reliability matter more here than raw feature count for this first release; a smaller platform that works reliably without internet is worth more to this audience than a feature-rich one that assumes a stable connection. Portability across low-end Android devices

is treated as equally important as portability across screen sizes, since the devices in use vary widely in age and specification.

## 5.5 Business Rules

- Only administrator-verified adults may register as mentors or opportunity providers.
- A student profile can only be created or edited by the student themselves or by a teacher registered at the same school.

**A mentor match cannot proceed to contact-sharing without explicit teacher or administrator approval**

<Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.>

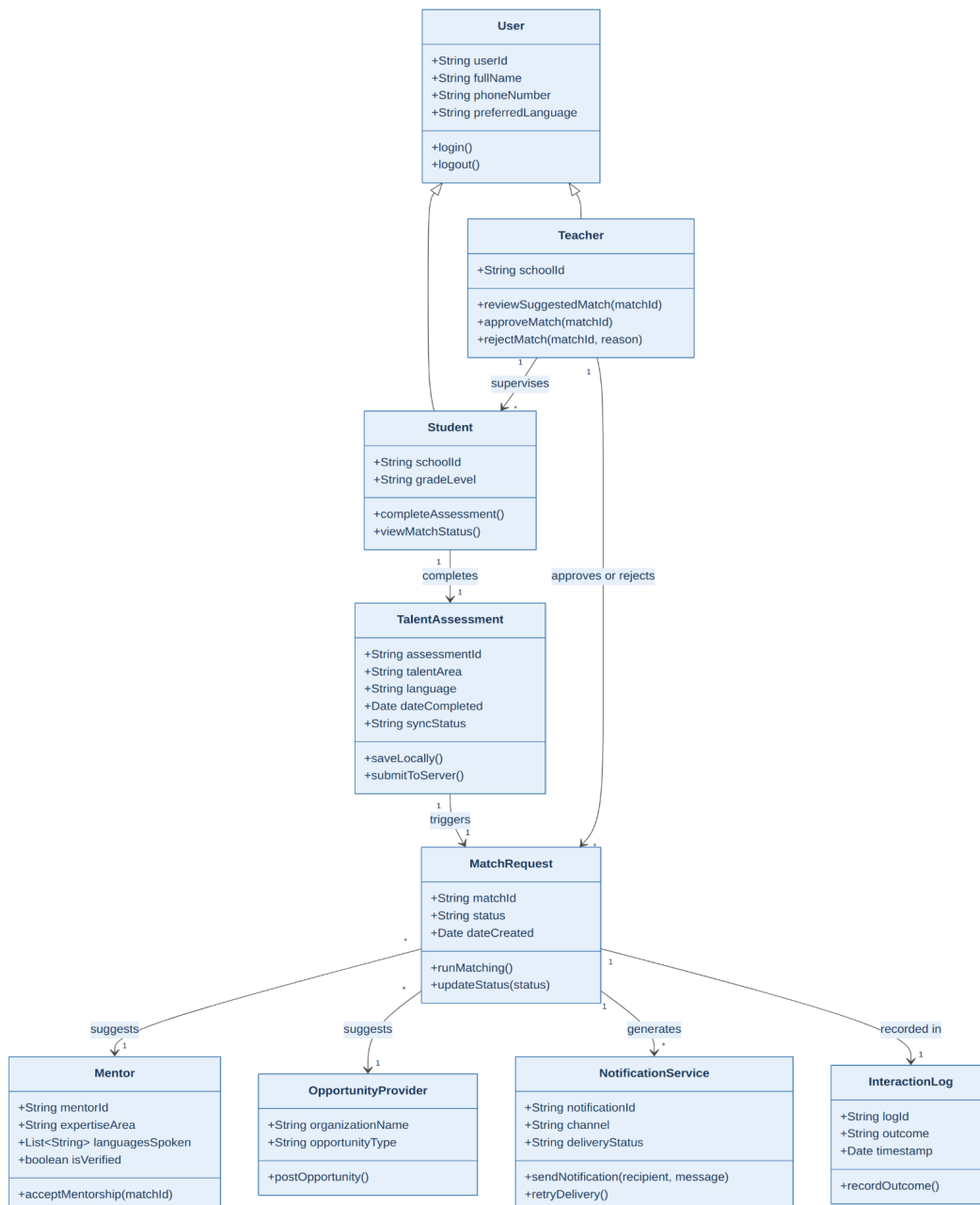
# 6. Appendix

## Appendix A: Glossary

- Talent Profile: A student's saved record of their talent-assessment results, basic details, and any confirmed matches.
- Offline-first: A design approach where core features work without an internet connection, syncing to the server only when one becomes available.
- Sync Queue: The local, on-device list of actions (such as a completed assessment) waiting to be sent to the server.
- Opportunity Provider: A verified school, NGO, company, or scholarship body that posts scholarships, competitions, or internships.
- REB: Rwanda Education Board, the government body responsible for education policy and school-based mentorship programmes in Rwanda.
- Sprint: A fixed, short work period, two weeks in this project, during which a specific set of features is built and reviewed.

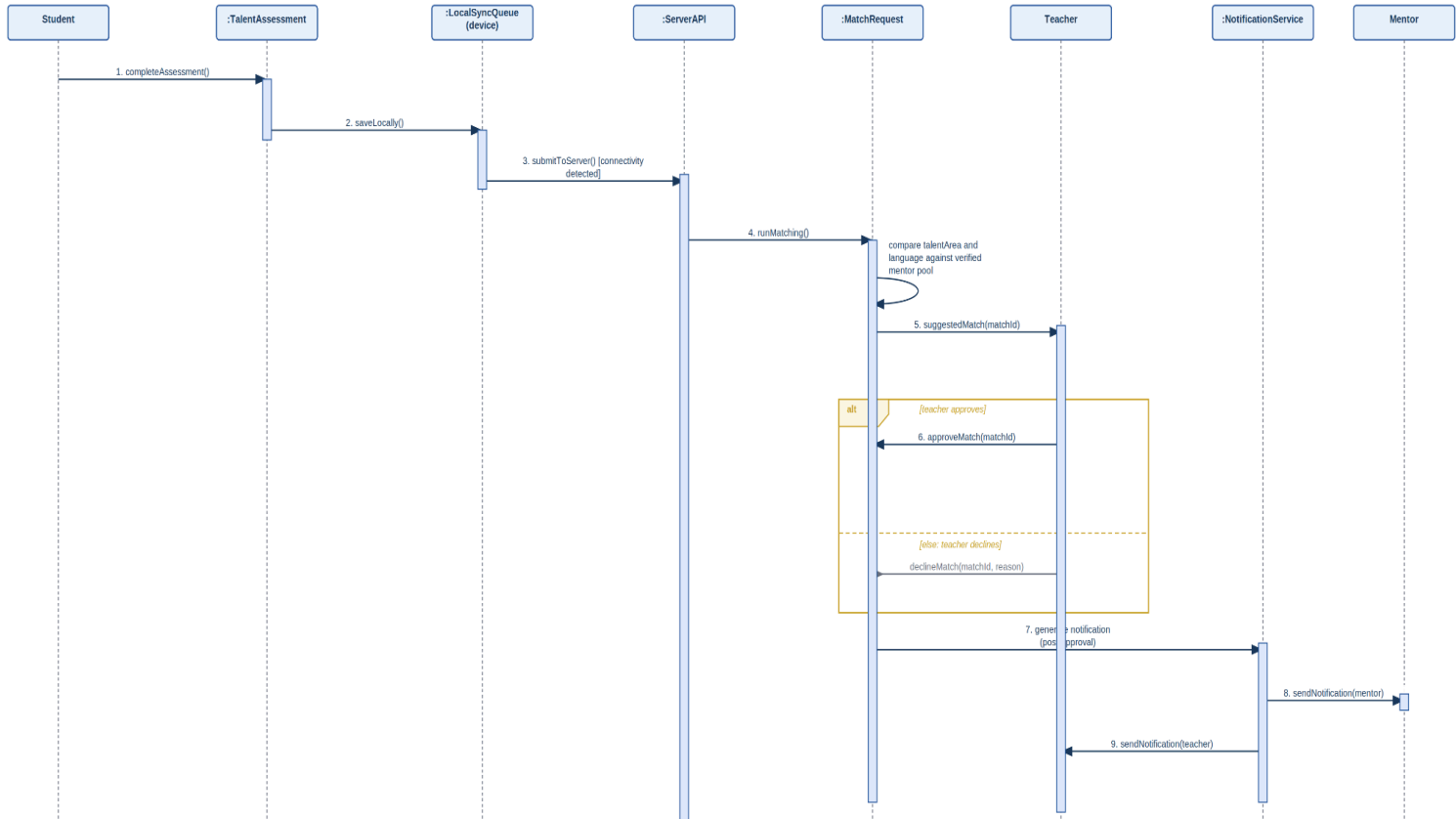
## Appendix B: Analysis Models

- Class Diagram:** My diagram focuses on the **User** superclass, and the subclasses are **Student** and **Teacher**, essential under generalization. Among the critical associations, the **Student** class "completes" the **TalentAssessment** class, which in turn "triggers" the **MatchRequest**. Then, the **MatchRequest** class "suggests" the list of **mentor** and "may suggest" a list of **OpportunityProvider**, making the **NotificationService** entries as a result of this action and being "recorded in" the **MatchAuditLog** class. Each class consists of its own attributes (i.e., **isVerified**, **syncStatus**) and operations (i.e. **runMatching ()**, **approveMatch ()**).



## 2. Sequence diagram

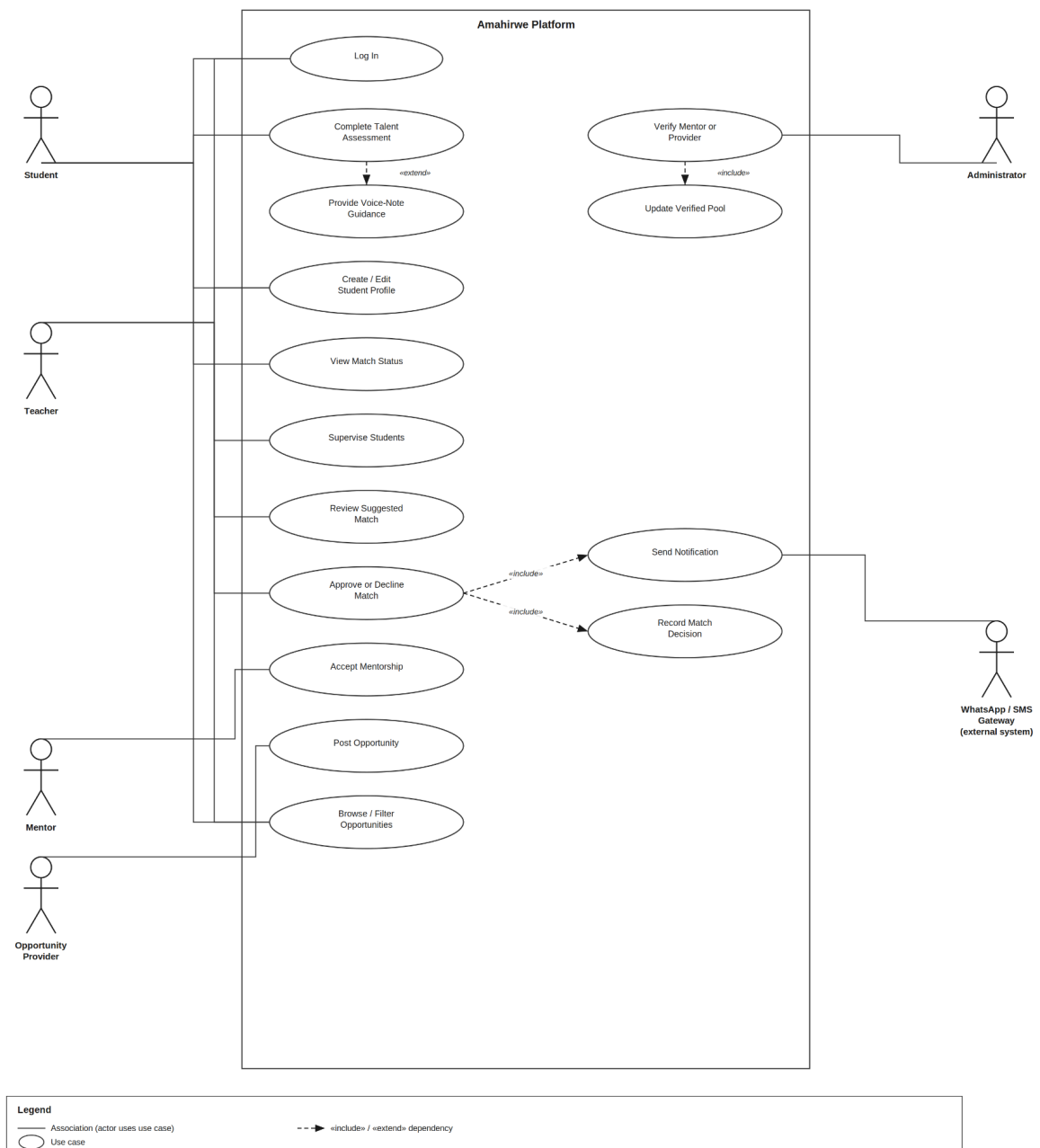
My focus in my diagram is to know the eight lifelines and their messages being delivered in sequence, including messages from **Student** to **:TalentAssessment** to **:LocalSyncQueue**. Then there is another message that comes from **:ServerAPI** to **:MatchRequest**. The activation bars tell us the object that is currently processing. The **alt** fragment is a core part of the diagram since it is splitting **:approveMatch()** and **:declineMatch()** into separate parts. The self-call loop on **:MatchRequest** tells us matching logic, and the final messages to **:NotificationService** show the broadcasting to mentor and teacher.



Match decision and delivery status are appended to MatchAuditLog by the server (omitted above for readability).

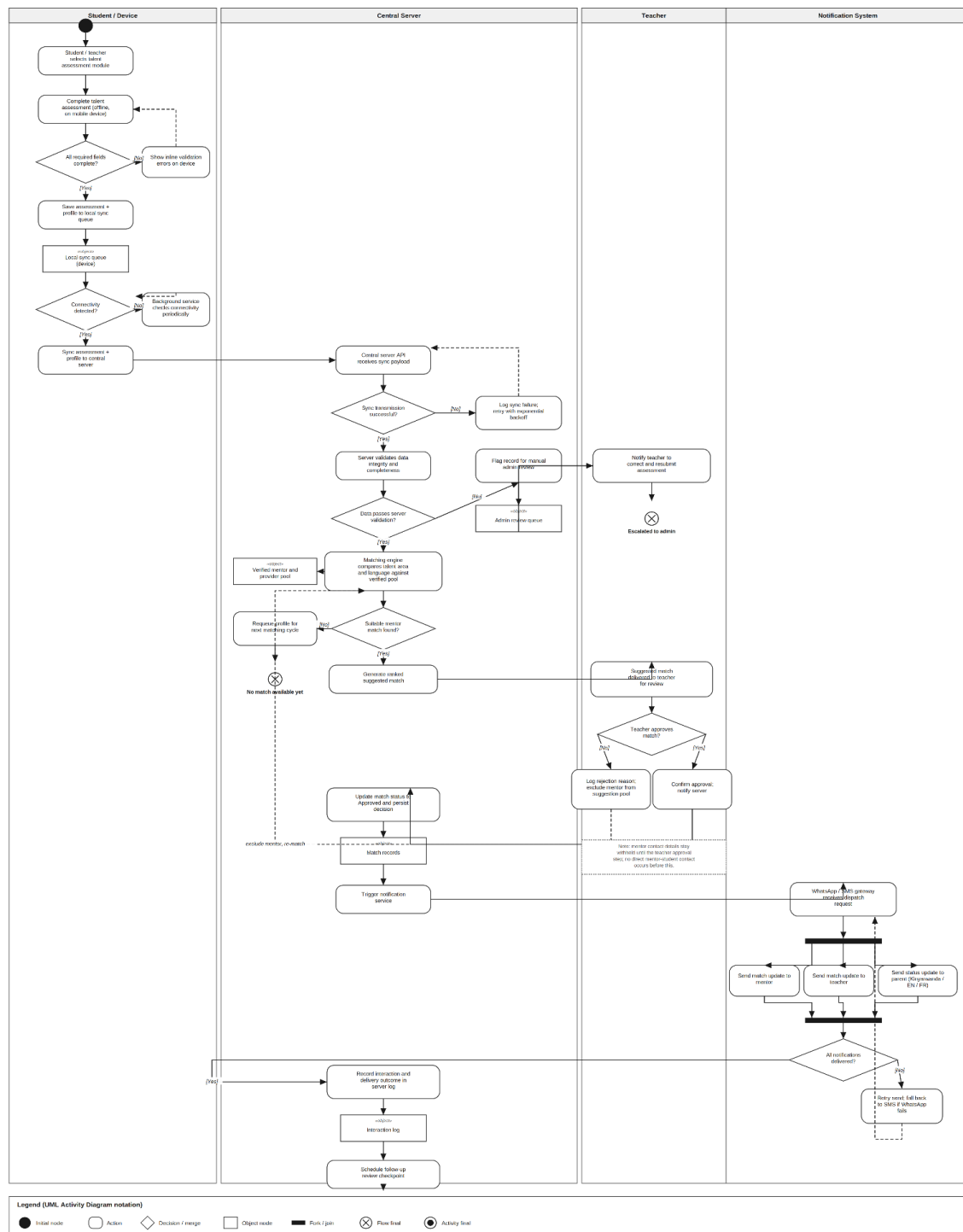
### 3. Use Case Diagram

The diagram's most important parts are six actors and their linkages to given use cases: **Student** has four connections, **Teacher** has five connections, and **Mentor**, **Provider**, **Administrator**, **WhatsApp/SMS Gateway** have one connection each. At this point, it is necessary to point out the two types of relationships: «**Include**» tells that the action **Approve or Decline Match** will activate with the actions **Send Notification** and **Record Match Decision**; «**extend**» shows **Provide Voice-Note Guidance** as optional, not mandatory behavior.



#### 4. Activity diagram

The diagram traces one talent assessment through the full pipeline. A student completes it offline, and it's held in a local sync queue until connectivity is detected, then batch-synced to the server API. The server runs data validation on the payload, and if it passes, the matching engine compares talent area and language against the verified mentor pool. A teacher approval gate sits before any match is confirmed; no mentor gets contact details without it, per the diagram's stated business rule. Once approved, the notification service dispatches via the WhatsApp/SMS gateway to the



mentor, teacher, and parent. Each failure point sync failure with exponential backoff, failed server-side validation, no suitable match, a rejected match exclusion, or failed notification delivery with SMS fallback routes through its own exception-handling branch rather than terminating the flow.